Jin Tao

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Selected Work Samples

Home Assistant Robot

Demo: youtu.be/3TWB9y0iEz0

- Lead development of a TurtleBot4 home assistant platform at ZEKI, integrating SLAM, navigation, autonomous exploration, semantic mapping, and an AI agent that translates user requests into robot actions.
- Implement a distributed onboard architecture: Raspberry Pi runs SLAM and navigation; NVIDIA Jetson Nano runs semantic mapping and the agent system.

TaoRobot: Autonomous Driving Full Stack

Code and demos: github.com/JinTTTT/taorobot

- Sole developer of an open-source ROS 2 autonomy stack, owning architecture, algorithm implementation, and integration in Gazebo simulation.
- Implemented pose-graph SLAM, particle-filter localization, A* path planning with path smoothing, and pure pursuit trajectory tracking. The repository includes demonstrations of mapping, localization, and goal-directed navigation.

Magnetic Joint-State Estimation

Code: GitHub repository: magnetic orientation estimation

- Proposed and developed a ball-joint orientation estimation system using a magnet and two low-cost three-axis magnetic sensors to estimate yaw, pitch, and roll.
- The implementation combines magnetic-field simulation, physical-model calibration, and a lookup-based estimator. Evaluated the system against an Xsens MTi-630 IMU; the magnetic sensors provide the measurements used during normal operation.

3D Object Pose Estimation with a Single RGB Camera

Code: GitHub repository: RGB-only object pose estimation

- Developed an OpenCV-based project for estimating 3D object pose using a single RGB camera.